

CSA1708-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 3 : Python Program for Water Jug Problem

Aim

To develop a Python program that measures a required amount of water using two jugs of given capacities.

Algorithm

- Read the capacity of jug A, the capacity of jug B and the goal amount.
- Start with both jugs empty (A = 0, B = 0).
- Repeat until either jug contains the goal amount.
- If jug A is empty, fill jug A completely.
- Otherwise, if jug B is full, empty jug B.
- Otherwise, pour water from jug A into jug B until jug A is empty or jug B is full.
- Print the amount of water in both jugs after every step.
- When the goal amount is reached, print "Goal reached".

Program

```
a = int(input("Enter capacity of jug A: "))
b = int(input("Enter capacity of jug B: "))
goal = int(input("Enter goal amount: "))

x = 0
y = 0

while x != goal and y != goal:
    if x == 0:
        x = a
    elif y == b:
        y = 0
    else:
        t = min(x, b - y)
        x = x - t
        y = y + t

    print("A =", x, "B =", y)

print("Goal reached")
```

Output

```
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/shaik/OneDrive/Documents/3.py =====
Enter capacity of jug A: 3
Enter capacity of jug B: 5
Enter goal amount: 2
A = 3 B = 0
A = 0 B = 3
A = 3 B = 3
A = 1 B = 5
A = 1 B = 0
A = 0 B = 1
A = 3 B = 1
A = 0 B = 4
A = 3 B = 4
A = 2 B = 5
Goal reached
>>> |
```

RESULT

Thus, the Water Jug Problem was solved successfully using Python.